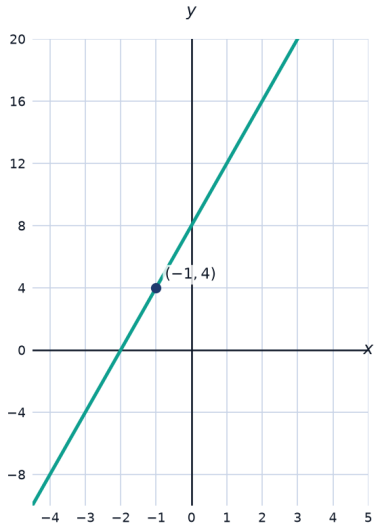
Quadratics & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

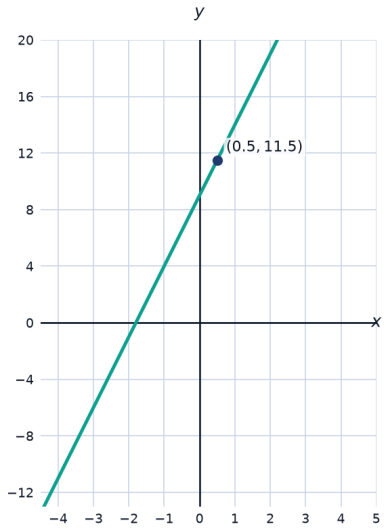
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(2)$.



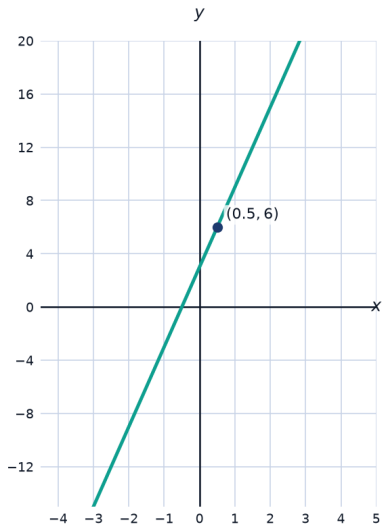
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for review: graphs and characteristics of quadratic functions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(2)$.

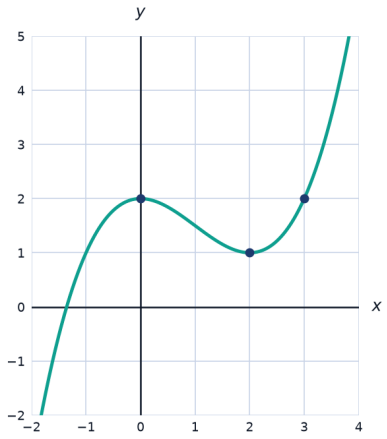


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for review: solving quadratic equations by factoring consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 4$ in complex numbers, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for complex numbers consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

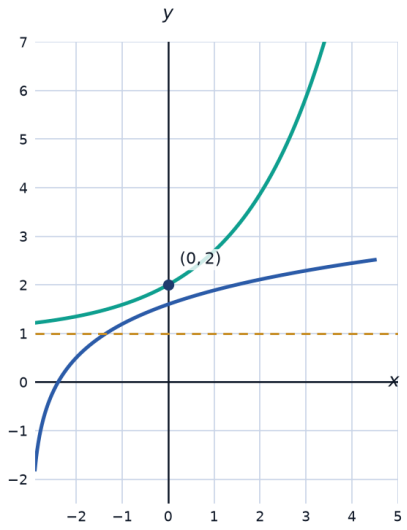
[Quadratics & Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve $x^2 - 4x + 13 = 0$ and express roots in the form $a \pm bi$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve $\sqrt{3x + 1} = x - 1$, checking for extraneous solutions.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Write $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ in standard form and identify the conic.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Sketch $y = \frac{2x+1}{x-3}$, showing all asymptotes and intercepts.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Conic Sections

Solve $2^{x+1} = 5^{x-1}$, giving your answer to 3 significant figures.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Trigonometric Functions

[Conic Sections · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability & Statistics

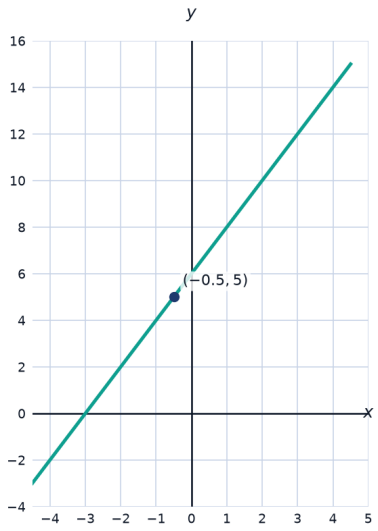
[Matrices & Systems · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(2)$.



a. 10

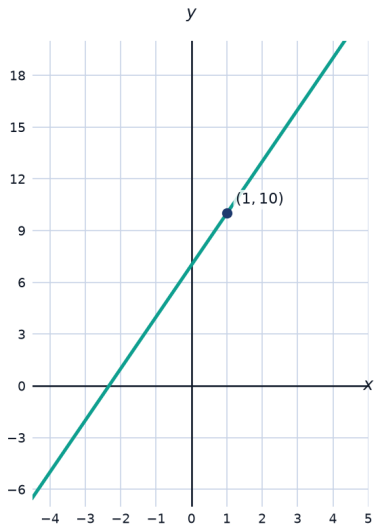
b. 12

c. 4

d. 14

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(2)$.



a. 13

b. 16

c. 6

d. 17

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: factoring techniques consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for factoring with sum and difference of cubes consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

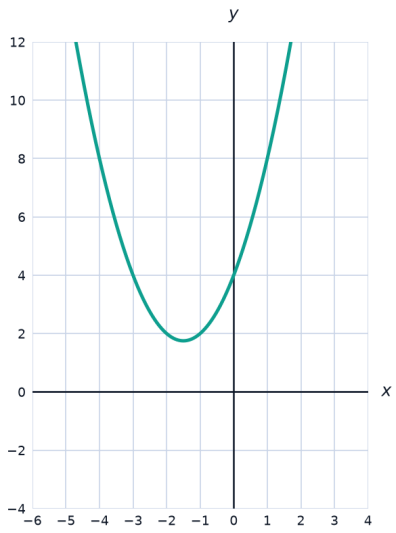
b. 17.5

c. 18.5

d. 20.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the nature of the roots of $x^2 + 3x + 4 = 0$.



a. Two distinct real roots

b. One repeated root

c. Four roots

d. Two non-real complex roots

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let m be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic equation $z^2 - (3 + 2i)z + (1 + 3i) = 0$. Let the roots of this equation be α and β. If the complex number w is defined such that $w = \frac{\alpha^2 + \beta^2}{2}$, find the modulus of w, $|w|$.

a. $\sqrt{5}$

b. $\frac{\sqrt{5}}{2}$

c. $\sqrt{29}$

d. $\frac{\sqrt{29}}{2}$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the quadratic equation $z^2 - (3 + i)z + (2 + 2i) = 0$, where z is a complex number. Let the roots of this equation be z_1 and z_2. Determine the value of $|z_1 - z_2|^2$.

a. 1

b. 2

c. 4

d. 5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the quadratic equation $2x^2 - (k - 1)x + (k + 1) = 0$ where k is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of k?

a. 7

b. 3

c. 10

d. 5

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following is equivalent to the expression $(3 + 4i)(2 - 5i)$, where $i = \sqrt{-1}$?

a. $26 - 7i$

b. $26 + 7i$

c. $-14 - 7i$

d. $6 - 20i$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $z^2 = \bar{z}$, find all solutions.

a. $z = 0$

b. $z = 0, 1, e^{\pm 2\pi i/3}$

c. $z = 0, 1$

d. $z = \pm 1$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Express $\frac{1 + \cos\theta + i\sin\theta}{1 + \cos\theta - i\sin\theta}$ in polar form.

a. $e^{i\theta}$

b. 1

c. $e^{2i\theta}$

d. $\cos\theta + i\sin\theta$ only for $\theta = 0$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the axis of symmetry and vertex of $y = -3x^2 + 12x - 1$.

a. Axis: $x = -2$; vertex $(-2, 11)$

b. Axis: $x = 2$; vertex $(2, 11)$

c. Axis: $x = 4$; vertex $(4, 1)$

d. Axis: $x = 2$; vertex $(2, -1)$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which complex number is its own conjugate?

a. $-i$

b. Any real number

c. $1 + i$

d. i

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the product of roots of $3x^2 - 7x + 2 = 0$.

a. 7

b. 2

c. $\frac{2}{3}$

d. $\frac{7}{3}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of roots of $3x^2 - 7x + 2 = 0$.

a. $\frac{2}{3}$

b. 3

c. 7

d. $\frac{7}{3}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A ball thrown upward has height $h(t) = -4t^2 + 16t + 5$. Find the maximum height.

a. 16

b. 5

c. 20

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

Find the image of the circle $|z| = 2$ under the map $w = z + 1/z$.

a. An ellipse

b. A line

c. A parabola

d. A circle of radius 3

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

Find the minimum value of $f(x) = x^2 - 6x + 10$.

a. 1

b. -1

c. 6

d. 10

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability & Statistics

A quadratic with roots $r_1 = 2$ and $r_2 = 5$ has equation:

a. $x^2 - 3x + 10 = 0$

b. $x^2 - 10x + 7 = 0$

c. $x^2 - 7x + 10 = 0$

d. $x^2 + 7x + 10 = 0$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Polynomial Functions

If $z_1 = 2\angle 45°$ and $z_2 = 3\angle 30°$ (polar form), find $|z_1 z_2|$.

a. $6\angle 75°$

b. 5

c. $\sqrt{13}$

d. 6

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Rational Expressions

Convert to standard form: $(3 + 4i)(1 - 2i)$.

a. $3 - 8i$

b. $11 - 2i$

c. $11 + 2i$

d. $7 + 2i$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the two square roots of -9.

a. ± 3

b. $\pm 3i$

c. $3i$ only

d. $\pm\sqrt{-9} = \pm 3$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Which is NOT a quadratic equation?

a. $2x^2 = 5$

b. $x^2 + x + 1 = 0$

c. $x^2 - 1 = 0$

d. $x^3 + 2x = 0$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

The discriminant of $2x^2 + 3x + 5$ is:

a. -31

b. -49

c. 31

d. 49

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